Name…… **Index No:** ……

Random Number:..... **Stream:**.....**Signature:**…….....

*Candidates should **NOT** write their Centre Name or Centre Number anywhere on this booklet*

545/2
CHEMISTRY
Paper 2
DEC 2020
2 hours

END OF TERM TWO EXAMINATIONS

Uganda Certificate of Education

CHEMISTRY

Paper 2

2 hours

INSTRUCTIONS TO THE CANDIDATES

Section **A** *consists of 10 – structured questions. Attempt* **all** *questions in this* Section.

Answers to Section **A** *must be filled in the spaces provided.*

Section **B** *consists of* 4 *semi – structured questions. Attempt any* **two** *questions from this* Section. *Answers to this* Section *must be written in the answer pages provided overleaf.*

In both Sections, *all your working must be clearly shown.*

1 mole of any gas occupies22,400 cm^3 at s.t.p.

*1 mole of any gas occupies24,000*cm^3 *at room temperature.*

[S = 32, N = 14, O = 16, C = 12, H = 1]

FOR EXAMINERS' USE ONLY

1	2	3	4	5	6	7	8	9	10	11	12	13	14	TOTAL

SECTION A: (50 MARKS)

1. (a) (a) Name the two **major** components of air. *(01 mark)*

……..

……..

(b) When a sample of dry air was bubbled through potassium hydroxide solution and the residual gas passed over strongly heated copper metal, a gas **G** was finally obtained.

(i) Identity **G**

……..

(ii) State why air was bubbled through potassium hydroxide solution. *(01 mark)*

……..

(c) Write equation;

(i) To illustrate your answer in (b) (ii). *(1½ marks)*

…….

……..

(ii) For the reaction of the residual gas with copper. *(1½ marks)*

……..

……..

2. Complete the following table *(04½marks)*

Mixture.	**Method of separation.**	**Principle used.**
Iodine and sodium chloride solution		
Potassium nitrate and potassium chlorate		
Liquid air		

3. Below is part of the periodic table. Use it to answer questions following. The letters used are not the correct symbols of the element.

I	II		III	IV	V	VI	VII	VIII
				S	**E**		**V**	
	Q		**R**					
P							**W**	

(a) (i) State the general name given to element represented by

(i) **P** …….. (ii) **Q** ……. *(01 mark)*

(b) Write the

(i) electronic configuration of **W**. *(0½ mark)*

……

(ii) the formula of the ion of **R**. *(0½ mark)*

.....

(ii) Write equation for the reaction between the peroxide of **P** and water. *(1½ marks)*

.....

.....

(c) Using the outermost energy level electrons only, draw diagrams to show how atoms of elements **S** and **V** combine to form a compound. *(1½ marks)*

4. (a) A hydrocarbon Y contains 85.7% carbon. Calculate its simplest formula (C = 12, H = 1)

(02marks)

……

……

……

……

……

(b) 0.224g of hydrocarbon **Y** occupied 96cm^3 at room temperature (1 mole of gas occupies 24dm^3 at room temperature)

(i) Calculate its molecular mass. (01mark)

……

……

……

……

(ii) hence its molecular formula (01mark)

……

……

……

(c) write the structural formula of Y. (01mark)

……

……

……

5 (a) Define an allotrope *(0½mark)*

……

……

……

(b)Graphite and diamond are some of the allotropes of carbon. Briefly describe a simple test to prove that the two are allotropes. *(1½marks)*

……

……

……

……

(c). Figure 1.11 below shows a burning charcoal burner. Use it to answer questions following.

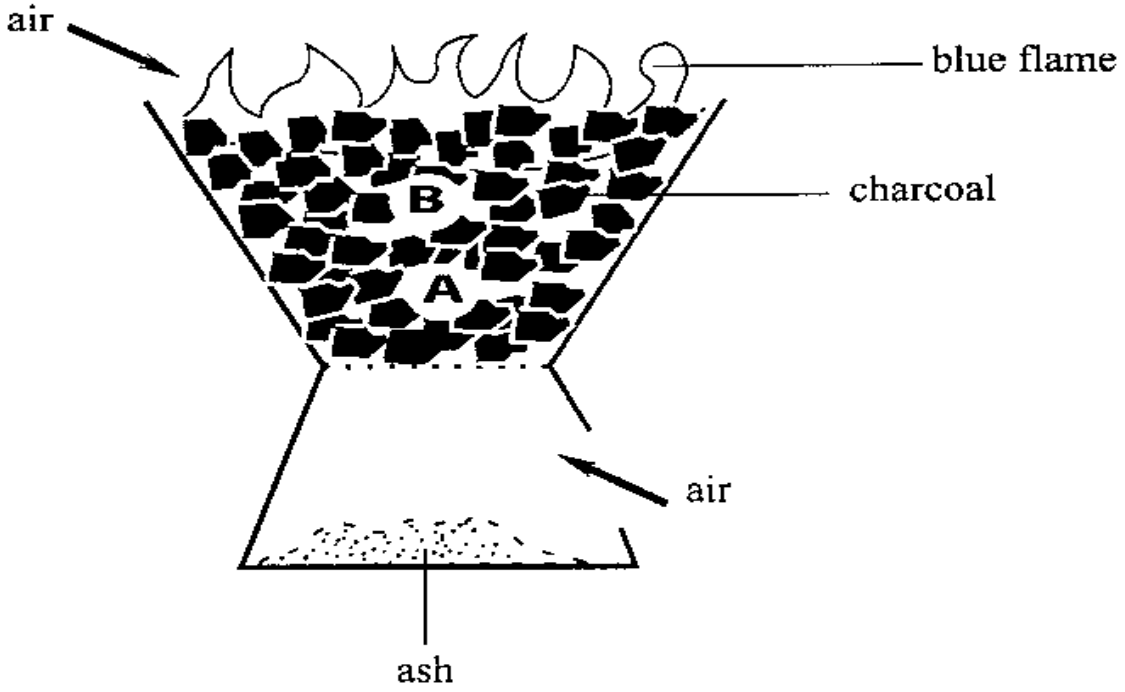


Fig 1.11 A burning charcoal burner

Write an equation for the reaction taking place at region

(i) **A** *(1½marks)*

……

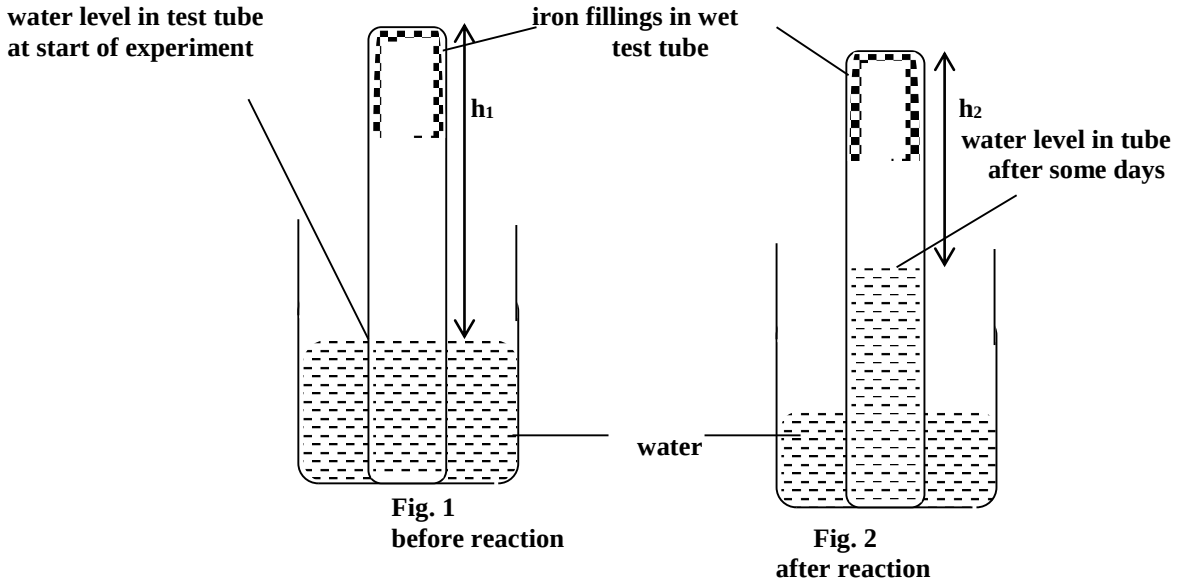
……

(ii) **B** *(1½marks)*

……

……

6. The diagrams in figure 1 and 2 below show a set up of an experiment which was performed to investigate how much air was used up when iron filings were kept in a wet test tube for some few days.



(a) (i) Name the process taking place above. *(½ mark)*

……

(b) Iron fillings are grey in colour.State what is observed after the reaction.

……

(c) (i) Why did the water level in the test tube change from h_1 to h_2?

……

……

(ii) State the approximate percentage of $h_1 - h_2$ (½ mark)

……

……

(d) State any conclusion that can be made from your answer in (c)(i) above . *(01 mark)*

……

……

(e) Name two ways of preventing the process named in (a) (i) *(01 mark)*

……

……

7. (a) State the condition(s) under which sodium can react with oxygen to form sodium peroxide. *(01mark)*

……

……

(b) Write equation for the reaction;

(i) leading to formation of sodium peroxide under the condition(s) which you have stated in (a). *(01½mrks)*

……

……

(ii) between sodium peroxide and water (1½marks)

……

……

(c) State the practical application of the reaction in (b)(ii) *(01mark)*

……

……

8. (a) During the laboratory preparation of sulphur dioxide gas, hydrochloric acid is reacted with compound **X**.

(i)Name compound **X.** *(0½ mark)*

.....

(ii)Write equation for the reaction leading to the formation of sulphur dioxide gas. *(1½ marks)*

.....

.....

(ii) State the condition(s) for the reaction in (a) (i). *(01 mark)*

.....

.....

.

(b) Name one reagent that can be used to identify the gas and state what would be observed if treated with the reagent named.

Reagent *(0½ mark)*

.....

.....

observations. *(01 mark)*

.....

.....

(ii) Explain your observations in (b) above *(01 mark)*

.....

.....

.....

9. (a) (a) Soap can be prepared from a vegetable oil or an animal fat.

What name is given to above process?

(01 mark)

……

(b) Briefly describe how soap can be prepared from simsim oil. *(03 marks)*

……

……

……

……

……

……

……

……

(c) State one advantage of soap over detergents *(01 mark)*

……

10. (a) State the particles that conduct electricity in

(i) An electrolyte *(0½mark)*

.....

(ii)An electrode *(0½mark)*

.....

(b) In electrolysis of copper(II) chloride solution using copper electrodes, state what was observed at

(i) Anode *(0½mark)*

.....

(ii) Cathode *(0½mark)*

.....

(c) Write an equation at

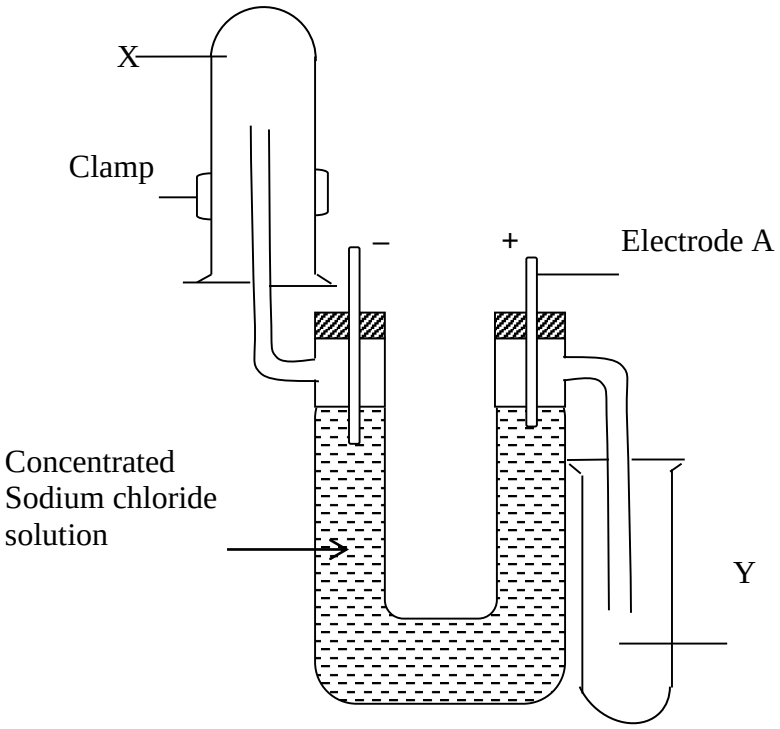
(i) Anode *(01 mark)*

.....

(ii)Cathode *(01mark)*

.....

(d)



The diagram above is U-tube voltammeter for the electrolysis of concentrated sodium chloride solution

(a) Identity

(i) Gas **X** (½ mark)

……

(ii) Gas **Y** (½ mark)

……

(iii) The material electrode **A** is made of (½ mark)

……..

SECTION B (*30 marks*)

Attempt any ***two*** questions from this section

11. (a) Nitrogen can react with hydrogen in the presence of a catalyst which is finely divided to form ammonia in the Haber process

(i)State the source of nitrogen and hydrogen *(01mark)*

(ii)Name the catalyst used in the reaction (0½ *mark*)

(iii)Explain why the catalyst is finally divided *(01mark)*

(iv)State two other factors which can affect the yield of ammonia in Haber process

(b) Write an equation for the reaction to show that ammonia can react with

(i) aqueous lead(II) nitrate *(1 ½ marks)*

(ii) Copper(II) oxide *(1 ½ marks)*

(c) When a mixture of ammonia and excess dry oxygen was passed over heated platinum wire catalyst, a colourless gas **P** was formed. On exposure to air, **P** formed a reddish brown gas **Q**

(i)Identify **P** and **Q** *(01mark)*

(ii)Write an equation of a reaction leading to the formation of **P** *(1 ½ marks)*

(d) Describe how **P** can be converted to nitric acid. (*06 marks)*

12. (a)Define the term enthalpy of combustion. *(01mark)*

(b) Describe using a well labelled diagram how the enthalpy of combustion of methanol can be determined in the laboratory. *(11 marks)*

(c) Methanol burns in oxygen according to the equation

$CH_3OH_{(l)} + {}^{3}/_{2}\, 0_2\,(g) \longrightarrow C0_2(g) + 2H_20_{(l)}\ \Delta H = -120KJmol^{-1}$

When a certain mass of methanol was burnt, the heat evolved raised the temperature of $100cm^3$ of water from 25.3^oC to 45.3^oC. Calculate the mass of methanol burnt. (3marks)
(Specific heat capacity of H_2O = $4.2Jg^{-1o}C^{-1}$; density of water = $1g/cm^3$, C= 12, H = 1,
O = 16) *(03 marks)*

13 (a) Define the term Rate of Reaction. ***(02 marks)***

b) State and explain how the rate of reaction between Zinc granules and 0.5M dilute sulphuric acid would be affected if;

(i) Zinc powder was used in place of Zinc granules. ***(01 ½ marks)***

(ii) 2M Sulphuric acid was used. ***(01 mark)***

c) Table below shows time taken for substance **Q** to react until completion under different

concentrations.

Time, **t** (s)	125	40	30	20	15
Concentration of **Q** ($moldm^{-3}$)	0.1	0.3	0.4	0.6	0.8
$\frac{1}{t}$ (S^{-1})					

(i) Copy and complete the table by computing the values of $\frac{1}{t}$ ***(2 ½ marks)***

(ii) Plot a graph of $\frac{1}{t}$ against concentration of **Q** on the graph paper provided. ***(05 marks)***

d) (i) From your graph, determine the time taken for the reaction to reach completion when

concentration is **0.5moldm^{-3}**. ***(1 ½ marks)***

(ii) Using the graph, deduce how the rate of reaction varies with concentration of **Q**.

(1 ½ marks)

14. (a) Sodium can be extracted from one of the common ores by electrolysis method. Calcium chloride is usually added to the electrolyte prior to electrolysis process; but the anode is normally separated from the cathode using iron gauze cylinder.

(i) Name the ore for sodium

(ii) state the role of calcium chloride.

(iii) State the condition under which the ore is electrolyzed.

(iv) State the material used to make the electrodes.

b) (i) Write equation for the reaction at the cathode.

(ii) Sodium formed during electrolysis is usually collected in dry nitrogen gas. Explain.

(iii) Name the by-product during sodium extraction.

c) State one use of;

(i) sodium

(ii) by-product in b(iii) above.

END.